



KIRLOSKAR CHILLERS PRIVATE LIMITED

A Kirloskar Group Company

Enriching Lives

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Engineering Specification

Product Name	R1234ze (E)
IUPAC Name	trans-1,3,3,3-Tetrafluoroprop-1-ene
Chemical Formula	$\text{CF}_3\text{CH}=\text{CHF}$
CAS Number	29118-24-9
Concentration (% w/w)	100
Molecular Weight (g/mol)	114.0 g/mol
Appearance	Clear
Color	Colorless gas
Odor	Slight, ether-like
Boiling point (1 atm)	-18.95 °C (-2.11 °F)
Liquid Density (kg/m ³)	1163.1, at 25 °C (77 °F)
Vapor Pressure (kPa)	498.5, at 25 °C (77 °F)
Vapor Density (kg/m ³)	26.32, at 25 °C (77 °F)
Critical Temperature	109.4 °C (228.9 °F)
Critical Pressure (kPa)	3636
Critical Density (kg/m ³)	489.0
Thermal Conductivity (Liq.)	0.0781 W/m.K, at 25 °C (77 °F)
Thermal Conductivity (Vapor)	0.0136 W/m.K, at 25 °C (77 °F)
Water Solubility	0.037%, @25 °C
Ozone Depletion Potential	0
Global Warming Potential	< 1
ASHRAE Std. 34 Safety Classification	A2L
Flammability Limits – ASTM E681-04 @ 21°C	Non Flammable
Flammability Limits – ASHRAE 34 @ 100°C	7% – 12% (by volume)